Supervised Learning

Feature Space $\mathcal{X}$

Label Space $\mathcal{Y}$

Goal: Construct a **predictor** $f : \mathcal{X} \to \mathcal{Y}$ to minimize

$$R(f) \equiv \mathbb{E}_{XY}\left[\text{loss}(Y, f(X))\right]$$

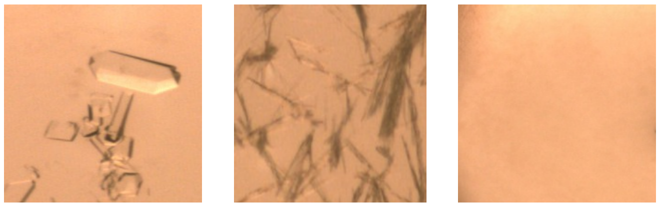
Optimal predictor (Bayes Rule) depends on unknown P_{XY}, so instead *learn* a good prediction rule from training data $\{(X_i, Y_i)\}_{i=1}^n \overset{\text{iid}}{\sim} P_{XY}$ (unknown)

Training data $\Rightarrow$ **Learning algorithm** $\Rightarrow$ Prediction rule

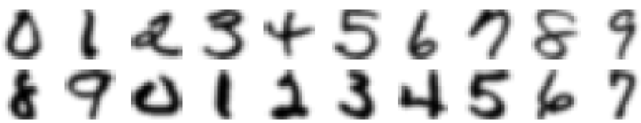
$\{(X_i, Y_i)\}_{i=1}^n$ $\qquad \widehat{f}_n$

Labeled

Training data



“Crystal” “Needle” “Empty”



“0” “1” “2” …

Human expert/
Special equipment/
Experiment

“Sports”
“News”
“Science”
…

Unlabeled data, X_i

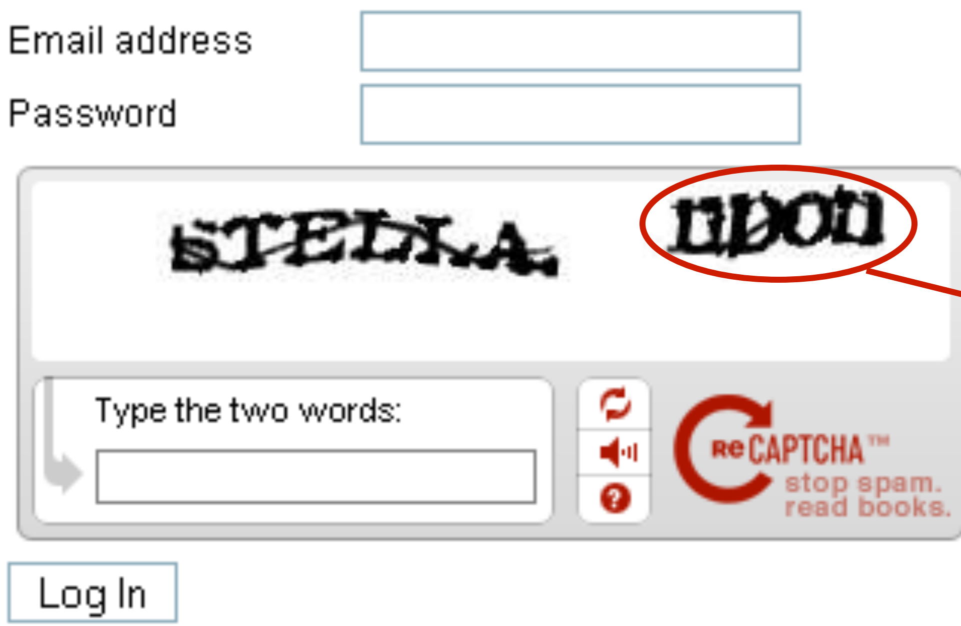
Labeled data, Y_i

Cheap and abundant !

Expensive and scarce !

Free-of-cost labels?

Luis von Ahn: Games with a purpose (ReCaptcha)



Word rejected by OCR
(Optical Character Recogintion)

You provide a free label!

Semi-Supervised learning

Training data $\Rightarrow$ **Learning algorithm** $\Rightarrow$ Prediction rule

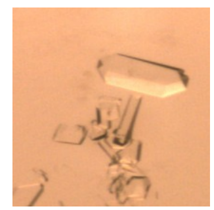
$\{(X_i, Y_i)\}_{i=1}^n$

$\{X_i\}_{i=1}^m$

$\widehat{f}_{n,m}$

Supervised learning (SL)

Labeled data $\{X_i, Y_i\}_{i=1}^n$



"Crystal"

X_i $\quad$ Y_i

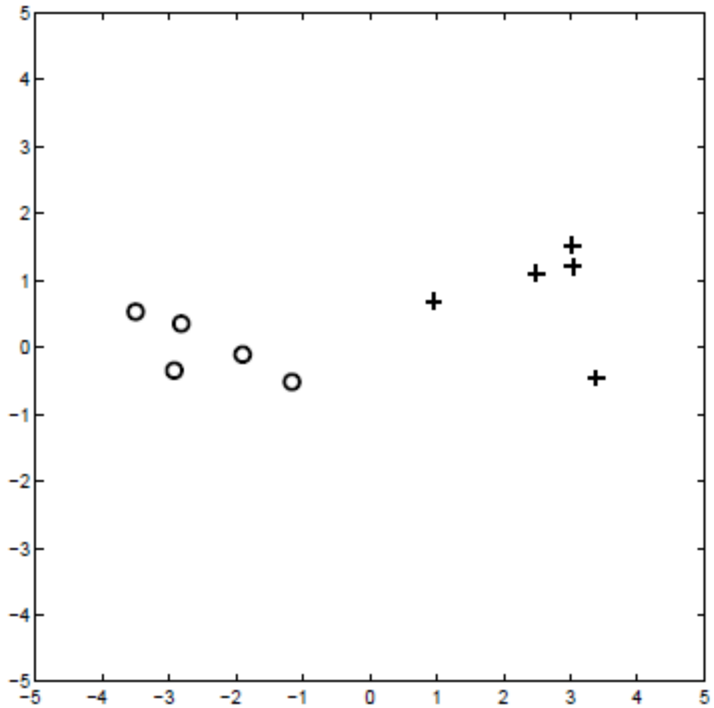
Semi-Supervised learning (SSL)

Labeled data $\{X_i, Y_i\}_{i=1}^n$ **and** Unlabeled data $\{X_i\}_{i=1}^m$ $\quad$ $m \gg n$

Goal: Learn a better prediction rule than based on labeled data alone.

Mixture Models

Labeled data (X_l, Y_l):



Assuming each class has a Gaussian distribution, what is the decision boundary?

Mixture Models

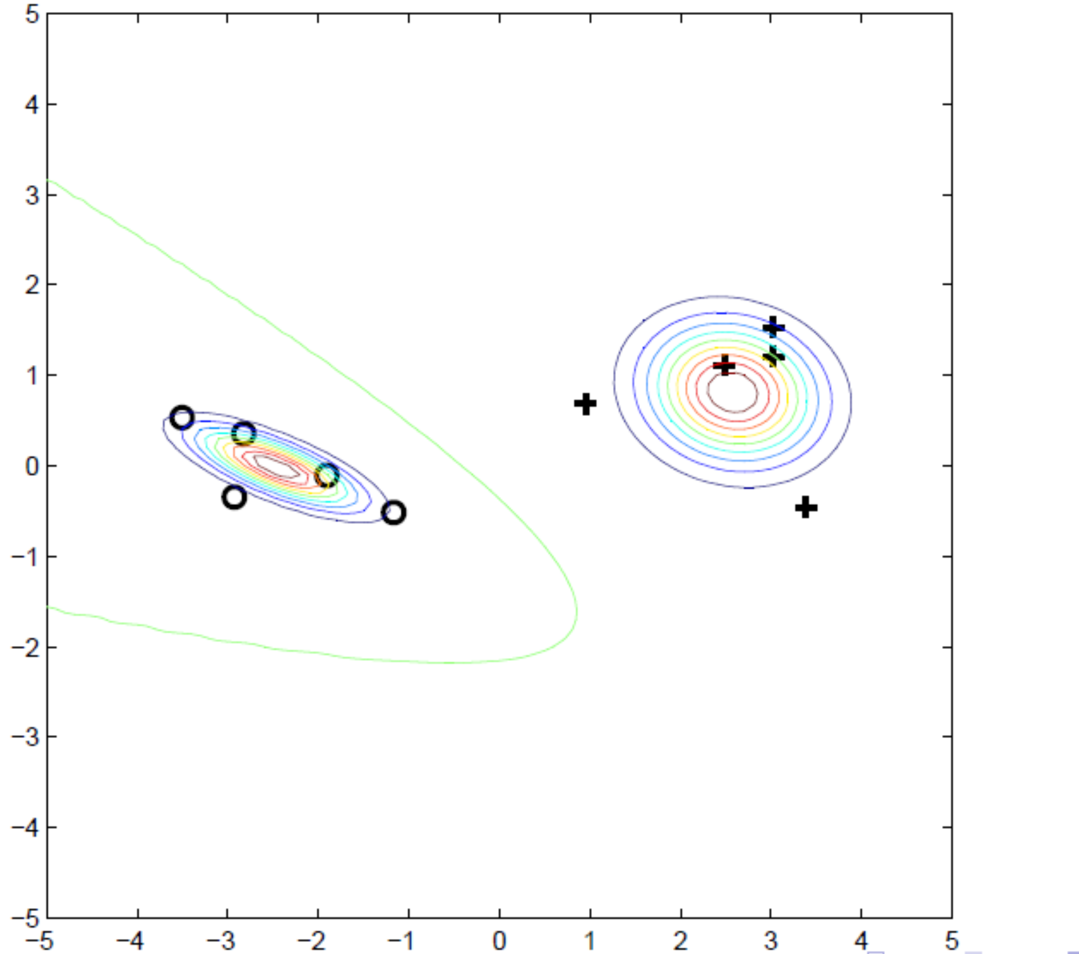
Model parameters: $\theta = \{w_1, w_2, \mu_1, \mu_2, \Sigma_1, \Sigma_2\}$
The GMM:

$$\begin{aligned} p(x, y|\theta) &= p(y|\theta)p(x|y, \theta) \\ &= w_y \mathcal{N}(x; \mu_y, \Sigma_y) \end{aligned}$$

Classification: $p(y|x, \theta) = \frac{p(x,y|\theta)}{\sum_{y'} p(x,y'|\theta)} \gtrless 1/2$

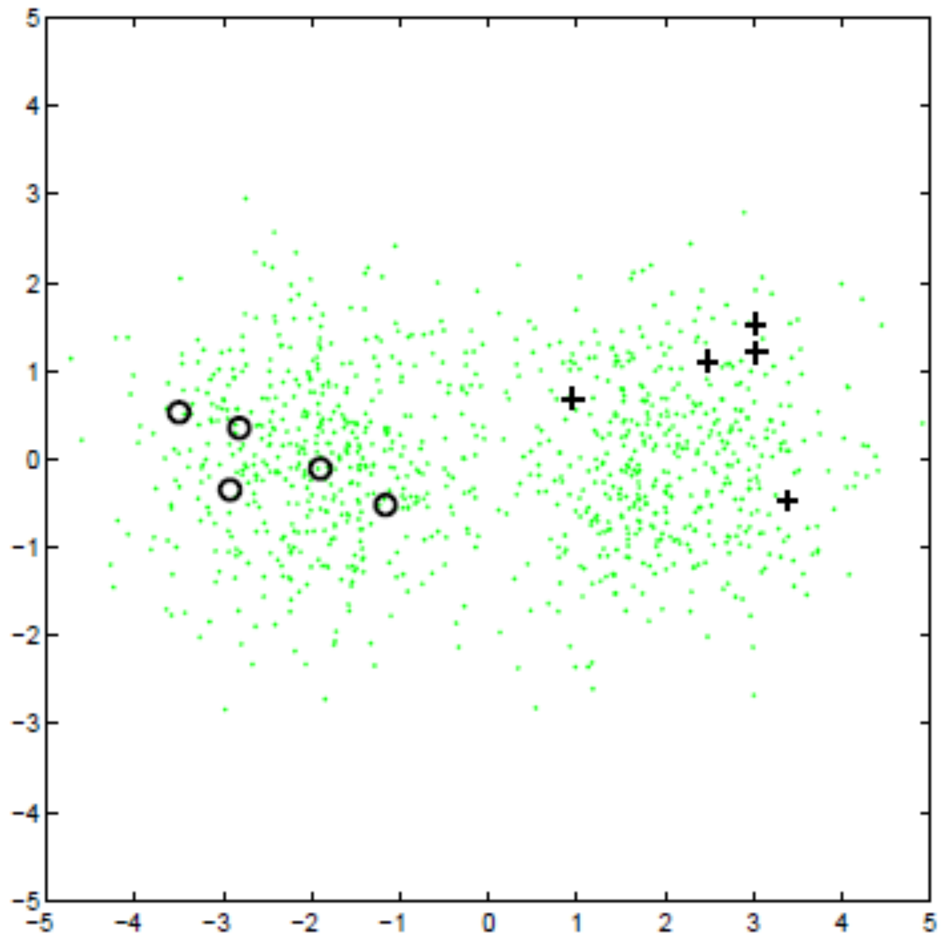
Mixture Models

The most likely model, and its decision boundary:



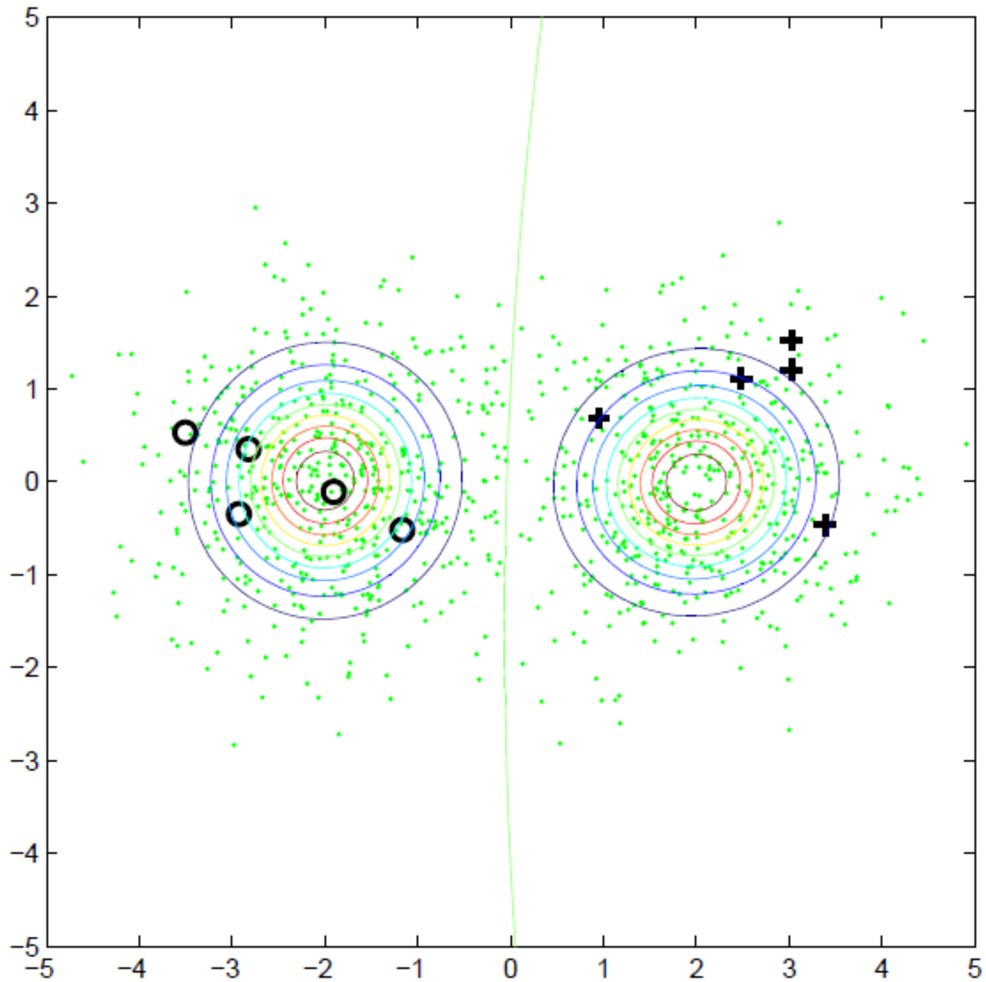
Mixture Models

Adding unlabeled data:



Mixture Models

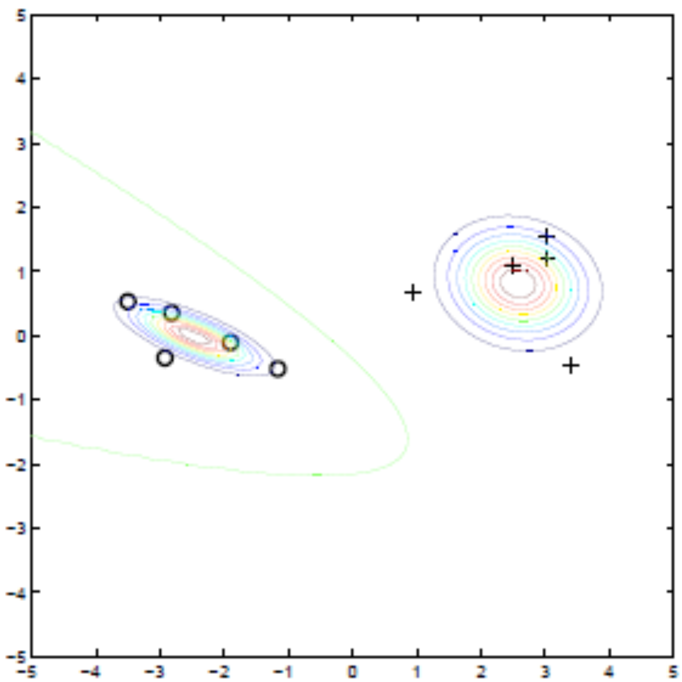
With unlabeled data, the most likely model and its decision boundary:



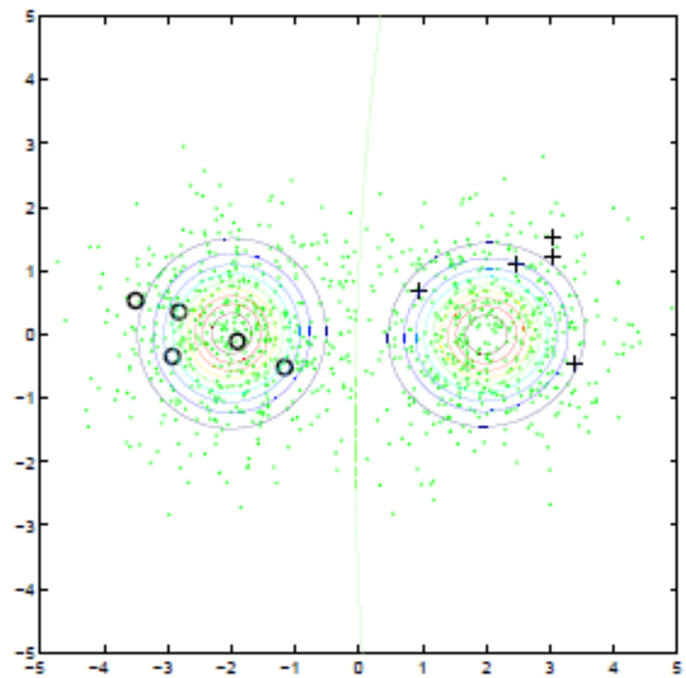
Mixture Models

They are different because they maximize different quantities.

$$p(X_l, Y_l|\theta)$$



$$p(X_l, Y_l, X_u|\theta)$$



Mixture Models

Assumption

knowledge of the model form $p(X, Y|\theta)$.

- joint and marginal likelihood

$$p(X_l, Y_l, X_u|\theta) = \sum_{Y_u} p(X_l, Y_l, X_u, Y_u|\theta)$$

- find the maximum likelihood estimate (MLE) of θ, the maximum a posteriori (MAP) estimate, or be Bayesian
- common mixture models used in semi-supervised learning:
  - Mixture of Gaussian distributions (GMM) – image classification
  - Mixture of multinomial distributions (Naïve Bayes) – text categorization
  - Hidden Markov Models (HMM) – speech recognition
- Learning via the Expectation-Maximization (EM) algorithm (Baum-Welch)

Gaussian Mixture Models

Binary classification with GMM using MLE.

- with only labeled data
  - $\log p(X_l, Y_l|\theta) = \sum_{i=1}^{l} \log p(y_i|\theta)p(x_i|y_i, \theta)$
  - MLE for θ trivial (sample mean and covariance)

- with both labeled and unlabeled data

$$\log p(X_l, Y_l, X_u|\theta) = \sum_{i=1}^{l} \log p(y_i|\theta)p(x_i|y_i, \theta) + \sum_{i=l+1}^{l+u} \log \left(\sum_{y=1}^{2} p(y|\theta)p(x_i|y, \theta) \right)$$

  - MLE harder (hidden variables): EM

EM for Gaussian Mixture Models

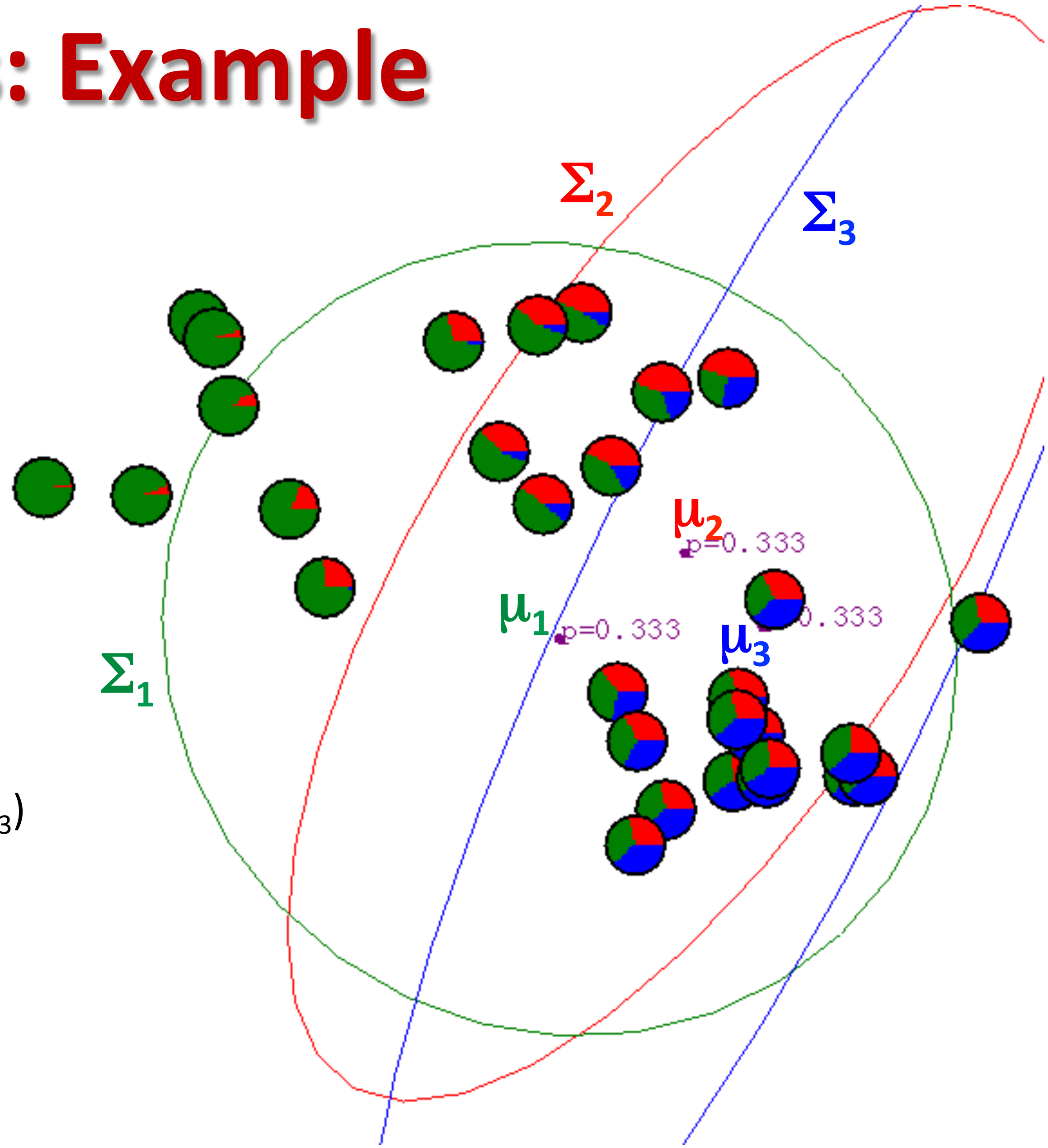
1. Start from MLE $\theta = \{w, \mu, \Sigma\}_{1:2}$ on (X_l, Y_l),
   - w_c=proportion of class c
   - μ_c=sample mean of class c
   - Σ_c=sample cov of class c

   repeat:

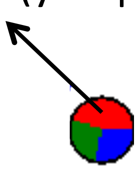
2. The E-step: compute the expected label $p(y|x,\theta) = \frac{p(x,y|\theta)}{\sum_{y'} p(x,y'|\theta)}$ for all $x \in X_u$
   - label $p(y=1|x,\theta)$-fraction of x with class 1
   - label $p(y=2|x,\theta)$-fraction of x with class 2

3. The M-step: update MLE θ with (now labeled) X_u

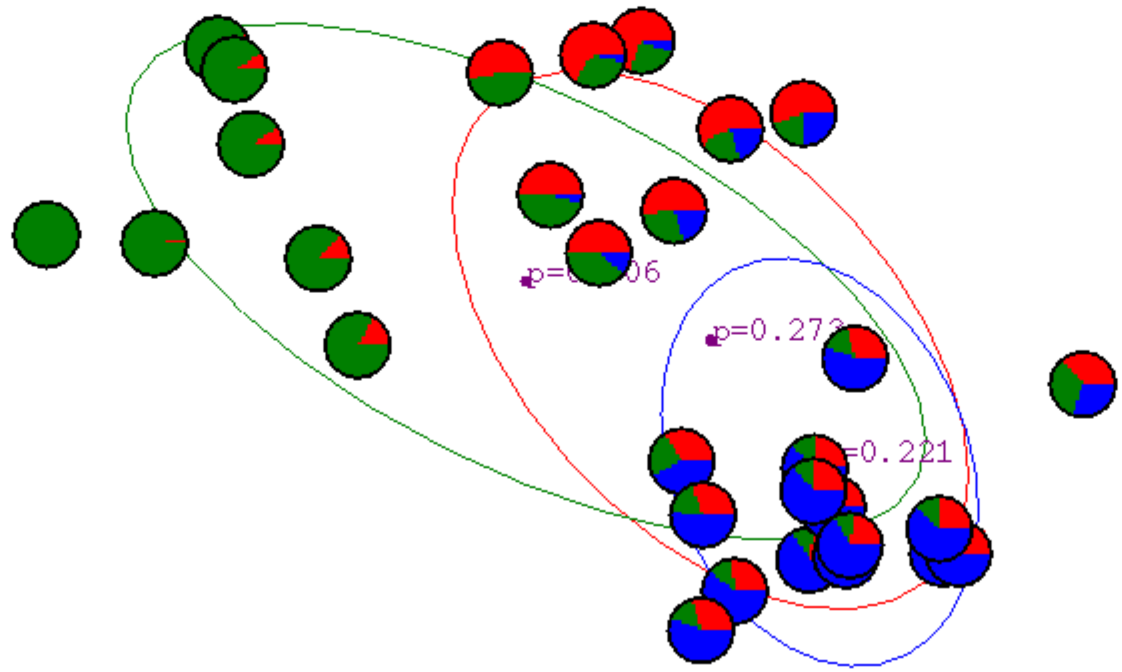
EM for GMMs: Example



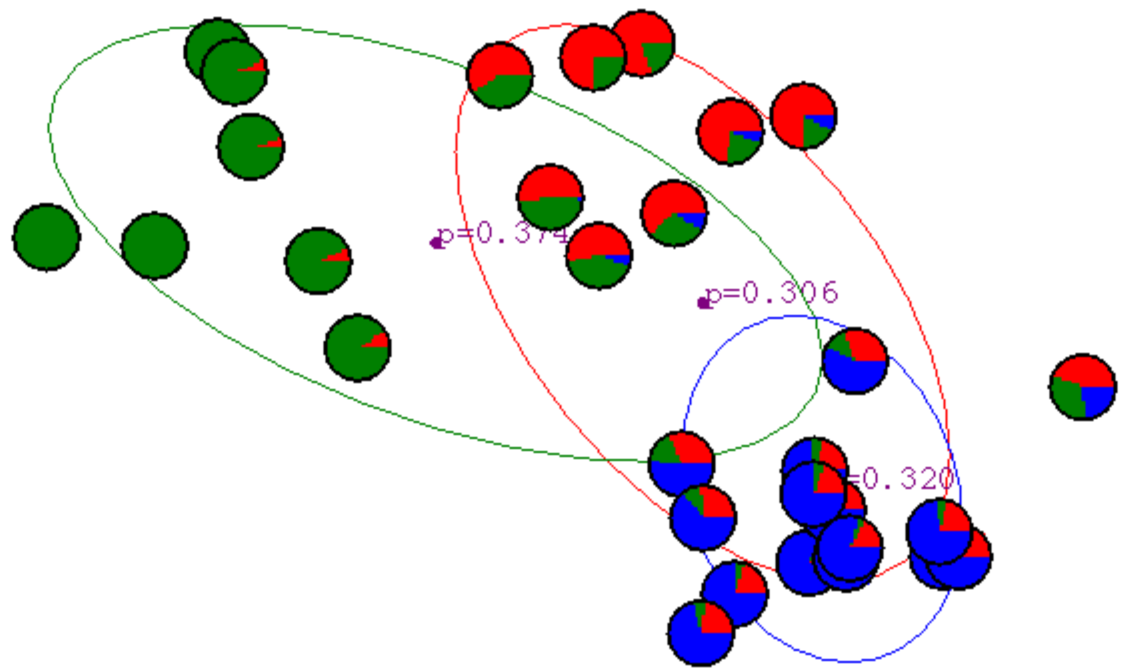
$P(y = \bullet \mid x_j, \mu_1, \mu_2, \mu_3, \Sigma_1, \Sigma_2, \Sigma_3, w_1, w_2, w_3)$



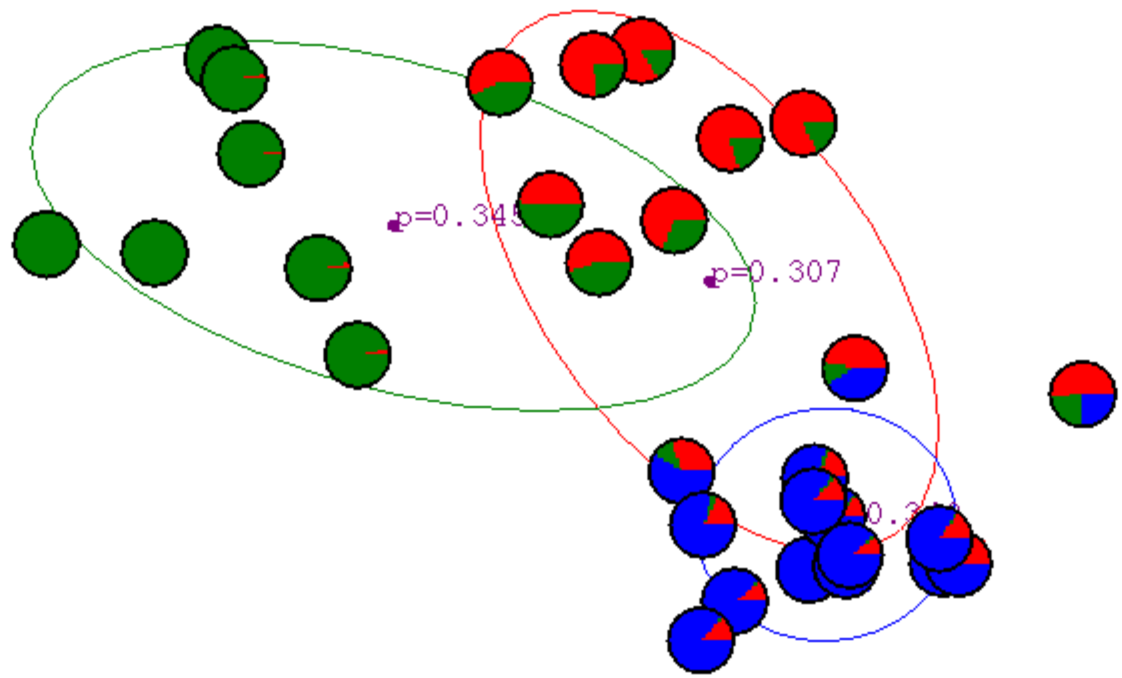
After 1st iteration



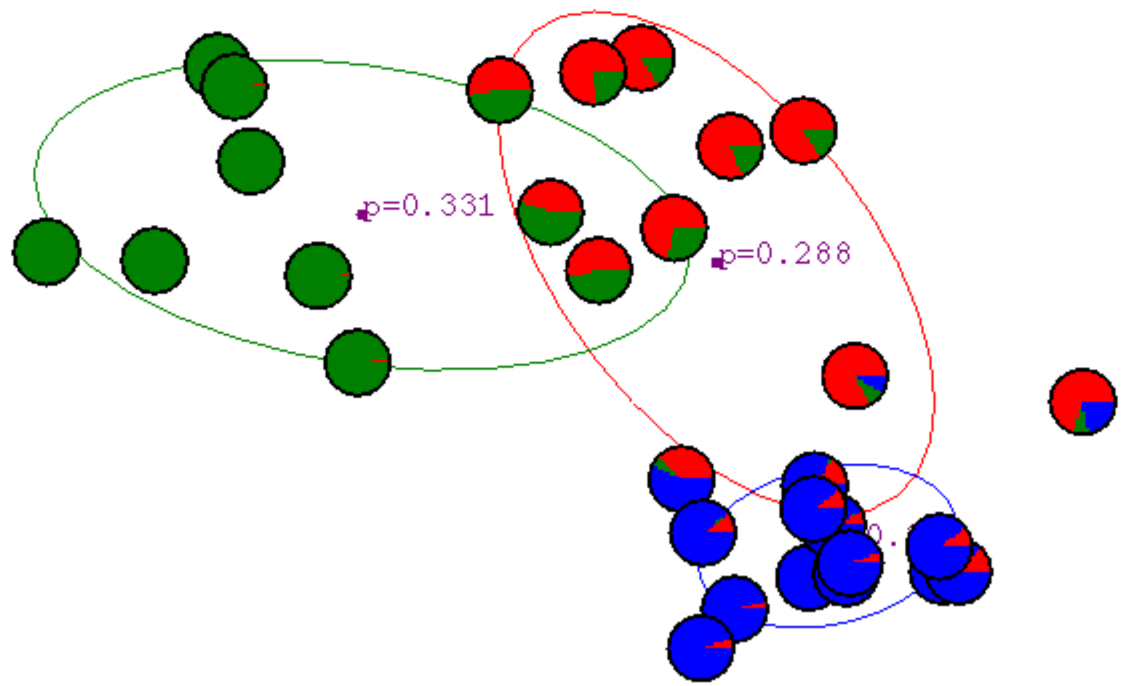
After 2nd iteration



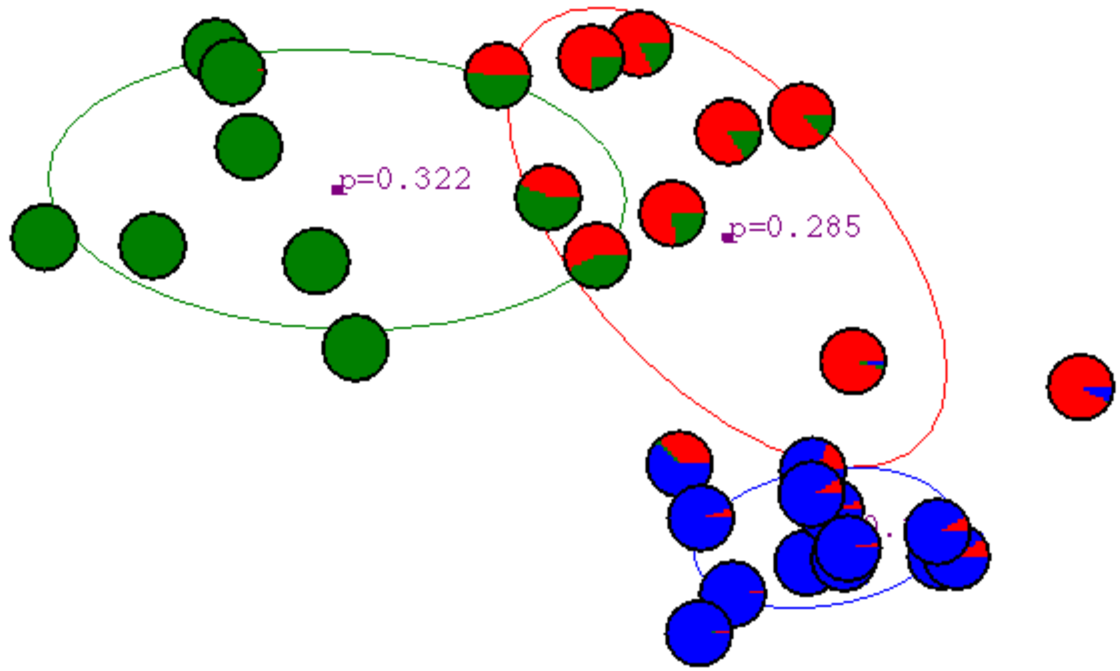
After 3rd iteration



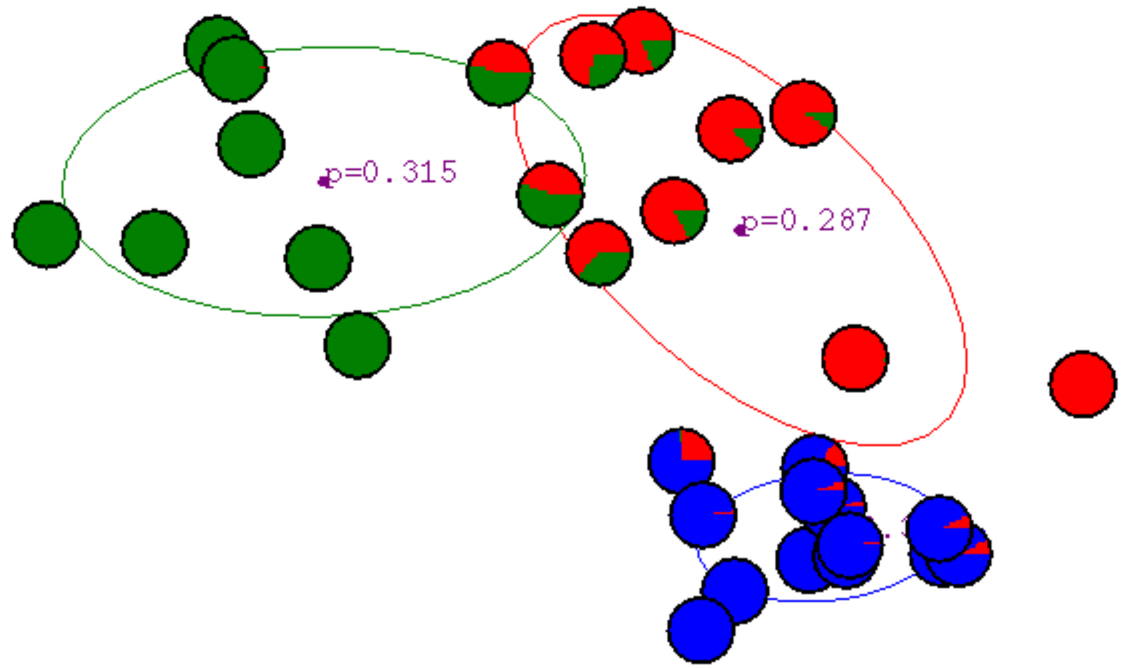
After 4th iteration



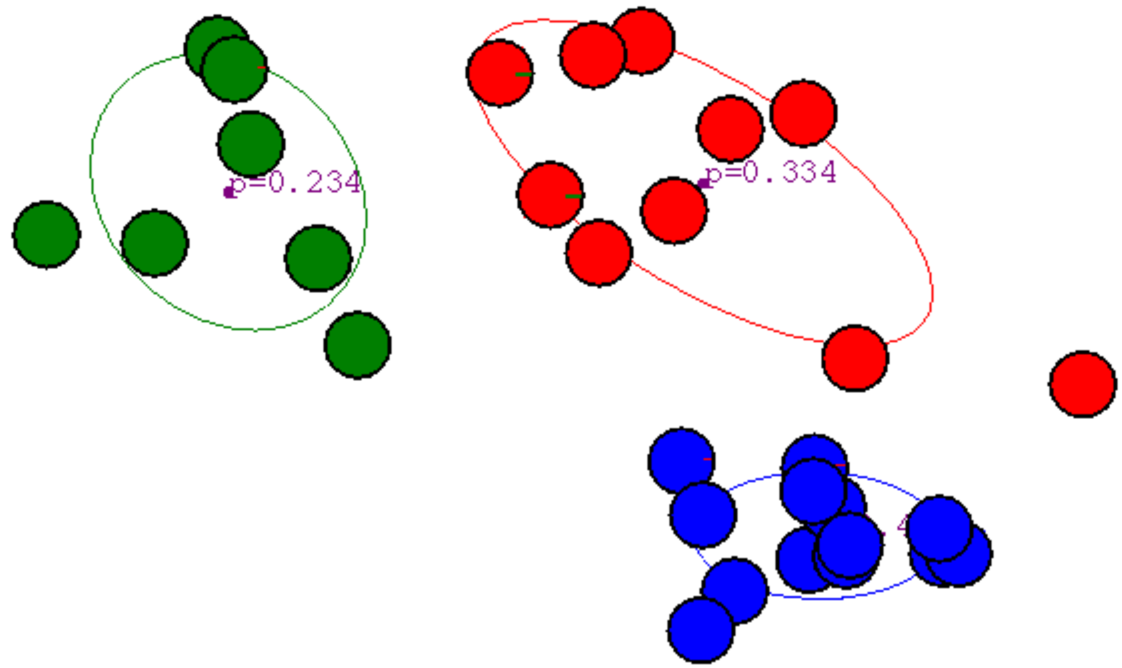
After 5th iteration



After 6th iteration

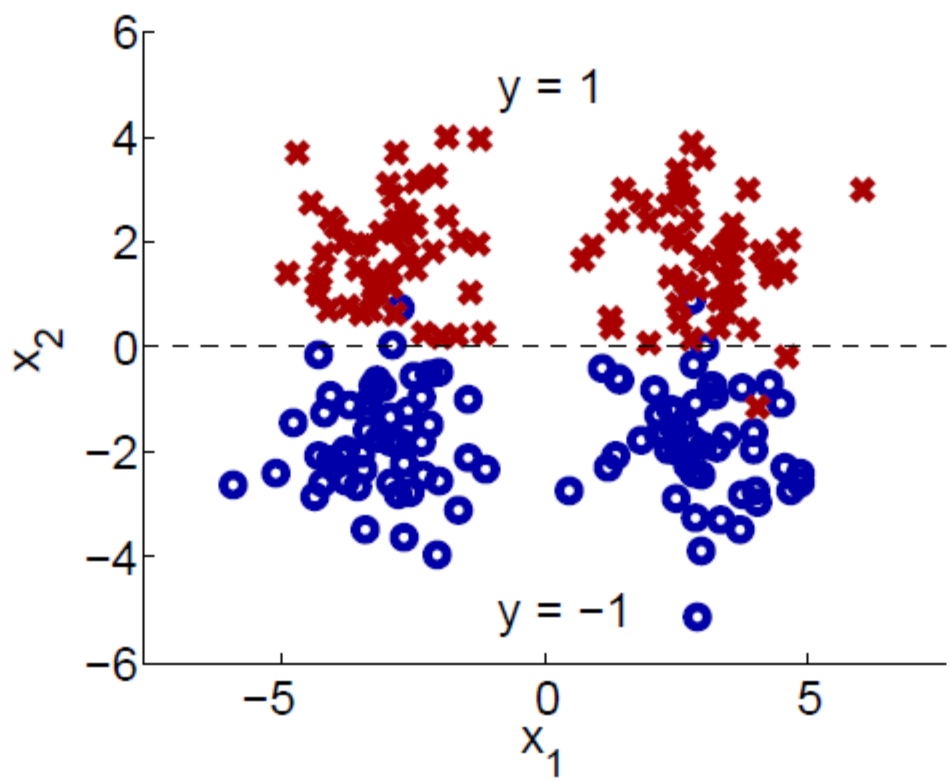


After 20th iteration

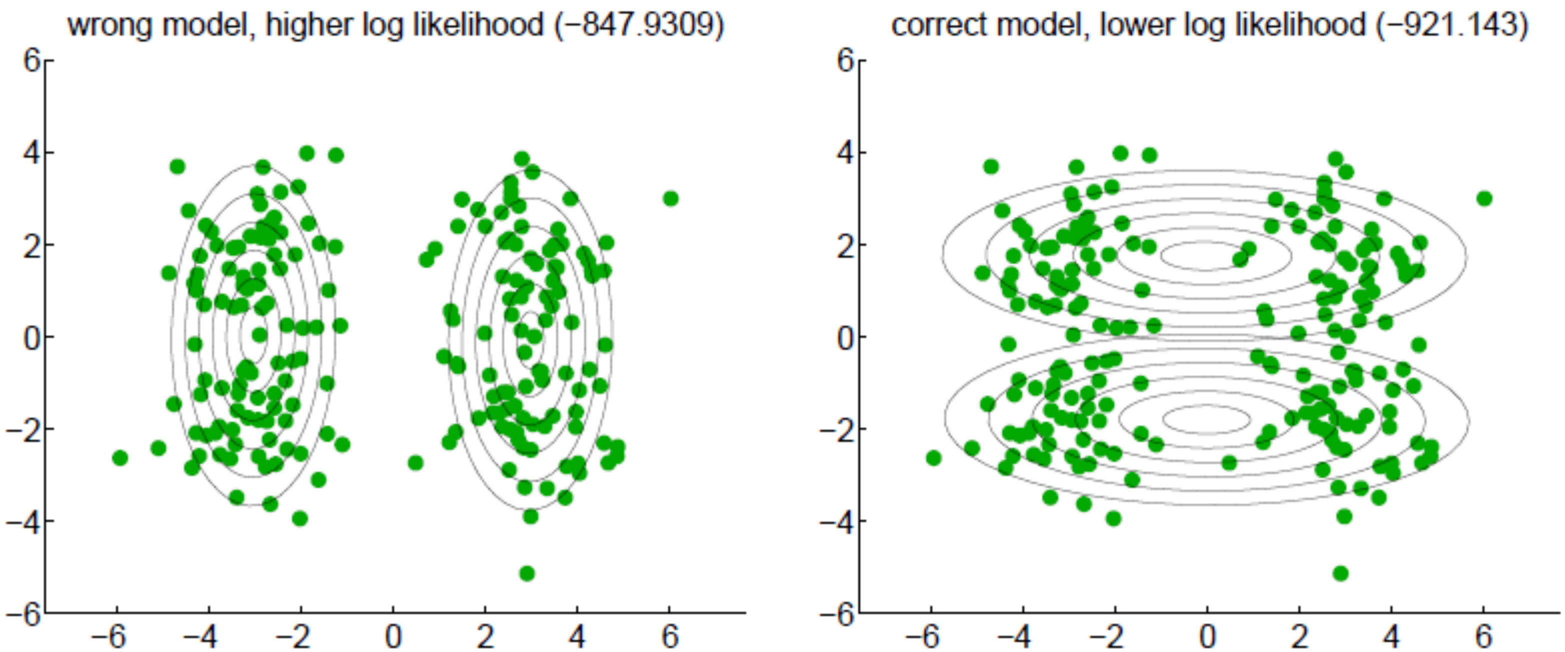


Assumption for GMMs

- **Assumption**: the data actually comes from the mixture model, where the number of components, prior $p(y)$, and conditional $p(\mathbf{x}|y)$ are all correct.
- When the assumption is wrong:



Assumption for GMMs



Assumption for GMMs

Heuristics to lessen the danger

- Carefully construct the generative model, e.g., multiple Gaussian distributions per class
- Down-weight the unlabeled data ($\lambda < 1$)

$$\log p(X_l, Y_l, X_u|\theta) = \sum_{i=1}^{l} \log p(y_i|\theta)p(x_i|y_i, \theta) + \lambda \sum_{i=l+1}^{l+u} \log \left(\sum_{y=1}^{2} p(y|\theta)p(x_i|y, \theta) \right)$$

Two views of an Instance

Example: named entity classification `Person` (Mr. Washington) or `Location` (Washington State)

instance 1: ... headquartered in (Washington State) ...
instance 2: ... (Mr. Washington), the vice president of ...

- a named entity has two views (subset of features) $\mathbf{x} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}]$
- the words of the entity is $\mathbf{x}^{(1)}$
- the context is $\mathbf{x}^{(2)}$

Two views of an Instance

instance 1:	. . . headquartered in (Washington State)^{L} . . .
instance 2:	. . . (Mr. Washington)^{P}, the vice president of . . .
test:	. . . (Robert Jordan), a partner at . . .
test:	. . . flew to (China) . . .

Two views of an Instance

With more unlabeled data

instance 1:	... headquartered in (Washington State)^L ...
instance 2:	... (Mr. Washington)^P, the vice president of ...
instance 3:	... headquartered in (Kazakhstan) ...
instance 4:	... flew to (Kazakhstan) ...
instance 5:	... (Mr. Smith), a partner at Steptoe & Johnson ...
test:	... (Robert Jordan), a partner at ...
test:	... flew to (China) ...

Co-training Algorithm

Blum & Mitchell'98

Input: labeled data $\{(\mathbf{x}_i, y_i)\}_{i=1}^{l}$, unlabeled data $\{\mathbf{x}_j\}_{j=l+1}^{l+u}$
each instance has two views $\mathbf{x}_i = [\mathbf{x}_i^{(1)}, \mathbf{x}_i^{(2)}]$,
and a learning speed k.

1. let $L_1 = L_2 = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_l, y_l)\}$.
2. Repeat until unlabeled data is used up:
3. Train view-1 $f^{(1)}$ from L_1, view-2 $f^{(2)}$ from L_2.
4. Classify unlabeled data with $f^{(1)}$ and $f^{(2)}$ separately.
5. Add $f^{(1)}$'s top k most-confident predictions $(\mathbf{x}, f^{(1)}(\mathbf{x}))$ to L_2.
Add $f^{(2)}$'s top k most-confident predictions $(\mathbf{x}, f^{(2)}(\mathbf{x}))$ to L_1.
Remove these from the unlabeled data.

Co-training

Assumptions

- feature split $x = [x^{(1)}; x^{(2)}]$ exists
- $x^{(1)}$ or $x^{(2)}$ alone is sufficient to train a good classifier
- $x^{(1)}$ and $x^{(2)}$ are conditionally independent given the class